Rohit Ramkumar

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Education

Vellore Institute of Technology (VIT), Vellore Expected 2028
B.Tech in Electronics and Communication Engineering (CGPA: 8.23/10) *Vellore, India*

Technical Skills

Languages: Verilog, Python, C/C++, 8051 & ARM7 Assembly, JavaScript
Hardware & EDA: Intel Quartus Prime, DE2-115 (Cyclone IV EP4CE115), Cadence Virtuoso (gpdk090), PIC32CM, ESP32, CC1101, nRF24L01
ML & Software: PyTorch, Gemini API (google-genai), React, FastAPI, NS-3, Matplotlib
Domains: FPGA/VLSI Design, Embedded Systems, RF/SDR, Cognitive Radio Networks, Applied Deep Learning

Projects

Cognitive Radio Network System | *NS-3, PyTorch, Gemini API* Research Project
- Architecting a cognitive radio network that combines NS-3 network simulation, PyTorch models, and a Gemini LLM-based decision engine.
- Migrating the I/Q pipeline from synthetic QPSK generation to physics-consistent I/Q derived from NS-3 `SpectrumWifiPhy`.
- Reviewed the NS-3 simulation codebase and identified multiple bugs, including a broken TCP retransmission tracker and a QPSK/OFDM physics inconsistency.
- Targeting patent filing and publication; funding being pursued.

EVOLVE: LLM-driven Verified Electronics Synthesis | *Gemini API, PyTorch, React, FastAPI*SENSE Design-a-thon; A
- Built an AI-powered circuit synthesis engine using a Gemini API and PyTorch deep-learning ensemble with hardware-grounded priors.
- Iterated the hardware architecture from AVR128DA48 to PIC32CM LS00 (BLE via RN4871), then to ESP32 with a Pre-Flash Comparator and Component Mixer.
- Developed the full-stack interface with React and FastAPI.

MAGMA: Magnetic-Gear and SMA Hybrid Actuator | *Python, Simulation* Honeywell Design-Thon
- Designed a hybrid actuator combining a coaxial magnetic gear with Nitinol (SMA) locking, driven by a BLDC motor.
- Wrote Python simulation scripts and authored the full design report.

Event-Driven Spectrum Sensing with Spiking Neurons | *Verilog, DE2-115, Quartus*NextGen-Innovathon 2026
- Building an FPGA spectrum sensing system based on LIF spiking neurons (Track 2: Emerging Tech & AI Hardware) in a team of three.

PRESENT-80 Cipher at Transistor Level | *Cadence Virtuoso, gpdk090* VLSI Lab Project
- Implementing the PRESENT-80 lightweight cipher S-box using static CMOS and pass-transistor logic; completed the 32-bit S-layer and progressing to the full cipher schematic.

Adaptive Noise Canceller | *Verilog, DE2-115, WM8731 Codec* Control Systems Project
- Implementing LMS-based adaptive FIR/IIR noise cancellation on the DE2-115 using the WM8731 audio codec.

Elevator Controller | *Verilog, DE2-115, Quartus* FPGA Project
- Designed a 7-state Moore FSM elevator controller with SCAN scheduling and an LCD driver; debugged on hardware.

Low-Cost 4-Channel SDR Prototype | *CC1101, nRF24L01* Hardware Project
- Built a low-cost four-channel software-defined radio prototype using CC1101 and nRF24L01 modules.

Additional Work

CMOS Inverter (Cadence, gpdk090): Designed and simulated an inverter (PMOS 285n/100n, NMOS 120n/100n); switching threshold 900 mV, $t_{pd} = 6.723$ ps, power 803.3 nW.
FPGA Basics: 4-tap FIR filter and clock divider in Verilog; interrupt-driven 8051 and ARM7 assembly programs.
Simulation Reports: Central Limit Theorem simulation (Matplotlib) for Random Processes; seminar report on Canny Edge Detection.